

2. Electronic food analyser. Immediate testing is important instead of sending the food items for analysis to expensive food testing labs and waiting for results. For this, we need electronic devices that we can use on our own to authenticate contaminated food products. OrganaDx (developed by MyDx, Inc., USA) acts as a preliminary tester before food products go to testing labs.

It detects harmful chemicals in food within minutes. The analyser offers one of the most practical food quality testing options for consumers today. It is a battery-operated handheld electronic analyser for consumers that contains a sensor to test for certain pesticides and helps to verify the safety of cannabis plants, fruits, and vegetables.

OrganaDx is the most practical food quality tester that allows detecting pesticides in food, ensuring that we do not unknowingly consume potentially dangerous cancer-causing pesticides at unsafe levels. It also allows testing of food quality for heavy metals to see whether it is safe for consumption.

The device is about the size of two iPhone 5s stacked together that fit into the palm of a hand. It has a small chamber where the sample is put to have a complete analysis of various chemicals in it. The analyser is connected via Bluetooth to a smartphone app, which people can use to report their data. The data will get shared (but information about the person who enters the data will be hidden) so that others can also benefit.

3. Electronic noses and tongues. Adulteration of food products due to use of pesticides, weedicides, and fertilisers during production, and use of additives on the way to consumers is one of the grave health risks. The concern for the quality of food products is relevant for producers, distributors, retailers, consumers, and national governments.

The electronic tongue and nose in the food industry authenticate tests of the different groups of food products. This is a rapid and powerful technique, which requires no special sample preparation to determine the quality of a product for the determination of food authenticity,

including adulteration. The use of these devices involves electronic products of advanced chemical and physical sciences combined with intuitive integration of microprocessors, advanced bioinformatics, and statistics.

The tests are based on voltammetric, bio-electronic, and piezoelectric platforms made from a variety of components. Electronic tongue or nose makes use of components based on the use of nanoparticles, enzyme biosensors, heavy metals, graphite-epoxy composites, metal oxide semiconductors, microelectrodes, microfluidic channels, pre-manufactured gas sensors, redox enzymes, and others as ideal resources for understanding and utilising their power in food science. The versatility of electronic devices lies with their capability to analyse multiple food items.

4. Blockchain and organic food. To ensure whether or not we are consuming authentic chemical-free food produce calls the need for a technology solution that can help resolve issues in one of the most effective, trusted, regulated, and consumer-driven food systems in the world. A new system to electronically certify organic products and ingredients has come into force, promising to tighten traceability, fight fraud, and collect reliable data on organic trade. The e-system (blockchain) tracks the movement of food products across the world. This aims to reduce fraud risks by providing greater transparency in the supply of organic products to consumers.

The blockchain was born with cryptocurrencies but is now growing in every sector where transactions take place. The blockchain is the technological basis of a platform in which those who participate in the supply chain interact with each other. It is an ecosystem in which from the suppliers and growers of organic seeds to commercial operators and certified processing companies, all are connected at different levels.

Blockchain—a distributed ledger technology offers the much-required traceability and transparency in an otherwise complex food supply chain by preventing fraudulence and ensur-

ing the authenticity of products to the consumers. Besides capturing data at every point within the supply chain ecosystem, a cloud and IoT powered blockchain system also ensures that it is shared amongst every supply chain participant such as farmer, manufacturer, supplier, distributor, and finally the consumer. The security of data is the immutability element of blockchain, which adds an additional level of security and prohibits any sort of fraudulent activity.

5. NIRS technology. The application of well-developed Near-Infrared Spectroscopy (NIRS) can be a rapid, cost-effective, and non-destructive method to verify the organic identity of food products. Recently, NIRS has been widely implemented in the food industry for analysing ingredients in foodstuffs and can be used without sample preparation. Spectral regions related to protein and fat content are among the most important ones for the classification model.

The models are based on data preprocessing using direct orthogonal signal correction (DOC), standard normal variant (SNV), and first and second derivatives provide the most successful results in terms of median sensitivity and median specificity.

6. Nuclear magnetic resonance (NMR) spectroscopy. Figuring out whether a fruit or vegetable has been grown under organic conditions is full of complications. The most reliable authentication technique currently used is based on analysing the stable isotope composition of nitrogen in food products, which is also not foolproof.

Scientists and engineers are looking towards the use of nuclear magnetic resonance spectroscopy to authenticate food products, including honey and olive oil. New methods with the potential to authenticate different organic foods based on the use of NMR technology are under development.

In conclusion, the development of fast, cost-effective, non-destructive, and portable electronic authentication techniques to certify the organic food products in the near future will be of great help for consumers to have healthy consumption of food and a safe environment. **EFY**